

Differential patterns of brain activity are correlated with sensitivity to morphological structure

Joanna Morris^{a,*}, Emma Kealey^{b,1}, Tess Rooney^{c,3}, Joemari Pulido^{d,3},
Natalia Alzate^{a,3}, Ethan Moore^{a,3}, Crismar Ramos-Marte^{a,3}, Frankie
Greene^{e,2}

^a*Department of Psychology, Providence College, Providence, RI, USA*

^b*Department of Linguistics, University of Southern California, Los Angeles, CA, USA*

^c*Columbia University School of Nursing, New York, NY, USA*

^d*Department of Cognitive, Linguistic & Psychological Sciences, Brown University,
Providence, RI, USA*

^e*Oregon Health & Science University and Portland State University School of Public
Health, Portland, OR, USA*

Abstract

Skilled reading requires us to rapidly retrieve a word's meaning from memory based on a brief exposure to a sequence of arbitrary, abstract visual symbols. Within the lexicon, the mapping from form to meaning takes place at the level of morphology, and rapid semantic retrieval is facilitated by lexical representations that are precise, specific, redundant and flexible. Given that both lexical quality and sensitivity to morphological structure are key components of reading success, can we identify patterns of brain activity that reflect individual differences in these characteristics? To answer this question, we collected event-related potential and response time data from par-

*Corresponding author: Joanna Morris, jmorris6@providence.edu

¹Work conducted while affiliated with the Department of Psychology, Providence College, Providence, RI, USA

²Work conducted while affiliated with the School of Cognitive Science, Hampshire College, Amherst, MA, USA

³These authors contributed equally to this work.

ticipants as they completed a lexical decision task featuring complex words that varied in the frequency of their morphological components. These data allowed us to determine individual differences in sensitivity to morphological structure. We also quantified individual differences in lexical quality (LQ) via estimates of vocabulary size and spelling performance, both of which are correlated with the precision and stability of lexical representations. We found differences between high LQ and low LQ participants in the N250 ERP component which has been hypothesised to reflect the processing sub-lexical orthographic units. These findings suggest that individual differences in lexical quality and sensitivity to morphological structure are reflected in distinct patterns of brain activity, and opens the avenue for further work to investigate the perceptual and cognitive processes that underlie these differences.

Keywords: morphology, event-related potentials, visual statistical learning, individual differences

1. Introduction

Reading can be defined as “the process of gaining meaning from print” (Rayner et al., 2001, p. 34). Because words are the primary vehicle for conveying meaning in language, learning to decode them is central to reading acquisition (e.g., Castles et al., 2018). Of course, we recognize that reading for comprehension involves much more than word recognition. Our knowledge of syntax tells us that “the man bit the dog” is newsworthy, while “the dog bit the man” is less so. Our understanding of social norms allows us to interpret the question “can you pass the salt?” as a request to do so, rather than a literal inquiry about physical ability. That said, skill in reading

comprehension rests to a considerable extent on knowledge of words.

Although all writing systems consist of abstract visual symbols that are mapped onto spoken language units, orthographies differ in terms of which spoken unit is mapped to a visual symbol: alphabetic systems map symbols to
15 phonemes, syllabaries to syllables, and logographies to words or morphemes.⁴ The linguistic unit targeted by a writing system has important consequences for how reading is acquired and processed.

Because alphabetic systems map visual symbols to individual speech sounds rather than to units of meaning, learning the relationships between
20 letters and sounds is necessary—and nonnegotiable—for beginning readers of any alphabetic script. However, the ease with which these mappings are learned depends on the orthographic depth of the language—that is, the consistency of the letter-to-sound correspondences. In a shallow orthography, one grapheme corresponds to one phoneme, whereas in a deep orthography,
25 the mappings are many-to-many rather than one-to-one.

English employs a deep orthography, in which grapheme-to-phoneme mappings are highly context-sensitive. They can vary based on letter position and the identity of neighboring graphemes. For example, at the syllabic level, post-vocalic consonants influence vowel pronunciation: the ⟨a⟩ in hat is pro-
30 nounced differently from the ⟨a⟩ in hate due to the addition of a final e—an example of a non-adjacent dependency. As a result, English spelling appears irregular at the level of individual graphemes and phonemes but shows

⁴Chinese is typically cited as the canonical example of a logographic system. Although it may have originated as such, in its current form it is more accurately classified as a morpho-syllabic system (Rayner et al., 2001).

greater regularity at the level of orthographic rimes,⁵ where common patterns can guide pronunciation (Treiman et al., 1995). Because both graphemes and
35 rimes are important for decoding, beginning readers must learn to recognize and associate both types of written units with their spoken counterparts.

In addition to its deep orthography, English also sacrifices phonological consistency in the service of morphological transparency for both stems and affixes. Stems often retain their spelling across derivations (e.g., heal →
40 health), and affixes contribute consistently to word meaning (e.g., help-less, home-less, worth-less, where -less = “without”). This prioritization of morphemic consistency introduces grapheme-to-morpheme regularities even when phonological mappings are irregular (Ziegler and Goswami, 2005; Rastle, 2019).

45 To become skilled readers of English, children must therefore develop a flexible word recognition system that accommodates mappings from symbols to phonemes, syllables, rimes, and morphemes. The context-sensitive, multi-level nature of English orthography means that phonological awareness, syllabic awareness, and morphological sensitivity each contribute to
50 reading development (Mahony et al., 2000).

Many of the relationships between graphemes, phonemes, syllables, and morphemes are not explicitly taught. Instead, readers must acquire this knowledge implicitly through statistical learning (SL)—the domain-general mechanism by which humans detect recurring patterns in the environment.
55 SL allows readers to internalize probabilistic correspondences between letters

⁵A rime is the part of a syllable consisting of its vowel and any consonant sounds that follow.

and sounds, as well as regularities among letters themselves. For example, orthographic units such as words (e.g., bird) or affixes (e.g., -er) are often defined by higher transitional probabilities between letters within a unit than between letters that span unit boundaries. The ability to detect and use such regularities—hereafter referred to as orthographic sensitivity—is crucial for
60 fluent word recognition and may emerge from SL mechanisms. Orthographic sensitivity follows a developmental trajectory throughout the elementary and middle school years (Treiman and Kessler, 2006) and is a robust predictor of reading ability. However, there is considerable individual variation in SL
65 performance among both children and adults (Arciuli and Simpson, 2012), and it remains unclear to what extent these differences contribute to variation in reading skill. Understanding how readers come to represent words in terms of their orthographic, phonological, and semantic components requires a closer look at the nature of lexical representations themselves.

70 1.1. *Lexical representation and lexical quality*

For literate speakers of a language, knowing a word involves familiarity with three distinct types of representation: a word’s sound, its meaning, and its written form. For skilled readers, all three are activated so quickly and seamlessly that they are perceived as a unified entity. This rapid, automatic,
75 and synchronous activation reflects the development of an “expert” mental lexicon—one characterized by high-quality lexical representations.

Perfetti (2007) defines lexical quality as “the extent to which the reader’s knowledge of a given word represents the word’s form and meaning constituents and knowledge of word use that combines meaning with pragmatic
80 features” (p. 359). High-quality representations are precise, specific, and flex-

ible (Perfetti, 2007). They include fully specified orthographic forms; phonological information that is both stored and recoverable from spelling; context-sensitive semantic representations; a complete morpho-syntactic specification; and strong integration across these components (Perfetti and Hart,
85 2002; Perfetti, 2007, 2017)

Even among skilled readers, substantial variability exists in both written language proficiency and in experimental measures of word recognition, such as response times and error rates in lexical decision tasks (Andrews, 2012). These differences may reflect variation in the precision of lexical representations. Andrews and colleagues (Andrews and Bond, 2009; Andrews
90 and Hersch, 2010; Andrews, 2012; Andrews and Lo, 2013) argue that lexical expertise is marked by precise representations that support rapid word identification with minimal activation of competing candidates. This efficiency reduces the need for attentional control and conserves cognitive resources for
95 higher-level processes such as inference and discourse comprehension.

Spelling ability is a particularly sensitive index of lexical quality, as it requires detailed and specific knowledge of a word’s orthographic form (Perfetti, 1992). In contrast, many reading tasks can be accomplished with only partial or approximate knowledge of how a word looks. As a result, it is
100 not uncommon to find skilled readers who struggle with spelling. This mismatch—fluent reading despite weak spelling—suggests a reliance on broad visual patterns and contextual guessing rather than detailed attention to orthographic structure (Frith, 2017). While such a strategy can support fluent decoding, it may fail to foster the robust, fully specified representations re-
105 quired for accurate written recall. Fine-tuning lexical precision requires more

than general reading experience; it depends on repeated exposure to specific words and a sensitivity to the regularities that link spelling to sound (Share, 1995).

Lexical precision can be indexed experimentally using the masked form
110 priming paradigm, in which responses to a word target (e.g., calm) are influenced by a briefly presented prime that shares orthographic similarity (e.g., clam). Such priming effects occur only for word targets; nonword targets do not benefit from formally similar primes under the same conditions, suggesting that the effect reflects pre-activation of lexical representations (Andrews
115 and Hersch, 2010; Forster and Davis, 1984). Importantly, the size of the priming effect varies with lexical precision. Readers with highly specified orthographic representations (e.g., for trail and trial) are less susceptible to activation by similar but mismatching primes (Forster and Taft, 1994). Consequently, smaller priming effects are taken as a marker of greater lexical
120 precision (Andrews and Hersch, 2010). If lexical quality hinges on the precision of orthographic representations, then individual differences in orthographic sensitivity—particularly in response to complex or morphologically rich words—may play a critical role in determining the efficiency of word recognition.

125 1.2. *Individual differences in complex word recognition*

Psychologists who study reading work within an experimental tradition in which they develop models based on the average responses of a sample of individuals to a manipulation of environmental variables (Cronbach, 1957). This approach rests on the implicit assumption that the cognitive architecture of
130 the reading system is uniform across skilled readers. But if readers differ in

the quality of their lexical representations, and if these differences lead to differences in the perceptual and cognitive processes underlying word reading, this assumption may not hold.

Andrews and Lo (2013) argue that although this assumption has been
135 useful in establishing the general principles of word reading, further progress
requires us to consider the role of individual differences. This approach has
already been shown to bear fruit in terms of providing clarity on a phe-
nomenon that has generated considerable discussion, morpho-orthographic
priming. Studies of morphological priming have consistently shown that
140 morphologically complex words facilitate lexical decision responses to their
stem morphemes across a range of languages, suggesting that morpheme
representations shared by primes and targets are activated early in the pro-
cess of word recognition. In the masked priming paradigm, a number of
studies have found that priming is not limited to primes that overlap in
145 meaning (e.g. ‘*payment*’-‘*pay*’), but also extend to pseudo-complex words
(e.g. ‘*corner*’-‘*corn*’). This evidence has been taken as support for ‘morpho-
orthographic decomposition’ accounts of morphological processing (?Rastle
and Davis, 2008) which assume an early orthographically-driven segmenta-
tion process that parses complex words into affixes and stems.

150 However, these findings are at odds with a meta-analysis of 16 studies
comparing masked priming for transparent and opaque words showed signif-
icantly stronger priming effect for transparent than opaque words (Feldman
et al., 2009). Moreover, equivalence in the degree of priming between se-
mantically transparent morphological relatives versus those that share a
155 semantically opaque or even pseudo-morphological relationship poses a prob-

lem for lexical models that rely on statistical learning (Plaut and Gonnerman, 2000; Baayen et al., 2011). These models capture sensitivity to morphological structure through learned associations between orthography and meaning; morphological relationships emerge as a consequence of these associations, and produce graded effects of morphological structure such that transparent
160 pairs, which share both orthographic and semantic features yield stronger priming than opaque pairs that share only orthography.

By examining individual differences between readers, Andrews and Lo (2013) were able to shed light on these discrepancies in the literature. If
165 morphology represents a learned sensitivity to the systematic relationships among the surface forms of words and their meanings, differences between English readers in their sensitivity to orthographic units and to the semantic structure of words could yield differences in the extent to which their lexicons become morphologically structured. Andrews and Lo (2013) found
170 that individual differences in in orthographic and semantic processing (indexed by scores on spelling and vocabulary tests respectively) were associated with systematic differences in participants' responses to morphological priming. Individuals with higher vocabulary than spelling scores ('semantic' profile) showed strong priming for transparent pairs but minimal priming for
175 opaque pairs. In contrast, individuals with higher spelling than vocabulary scores ('orthographic profile' indicating precise orthographic representations of words), showed equivalent priming for opaque and transparent pairs (Andrews and Lo, 2013).

Andrews and Lo (2013) argue that a 'semantic' reading profile may reflect
180 a holistic and contextually-driven reading strategy with greater sensitivity to

conceptual and semantic representations than to orthographic patterns. In contrast, an ‘orthographic’ profile may be associated with a bottom-up reading strategy focused on extracting frequently occurring orthographic patterns such as common affixes, regardless of their morphological status a process
185 that has come to be known as morpho-orthographic decomposition.

Rastle and Davis (2008) have proposed that among developing readers, morpho-orthographic decomposition is driven by statistical learning, i.e., sensitivity to the statistical structure of letter sequences. In particular, readers learn to mark low probability sequences as containing boundaries and
190 to group high-probability sequences into single units. If so, individual differences in sensitivity to statistical patterns may lead to differences in the strength and precision of morphemic representations, sensitivity to the morphological structure of words, and the use of such structures in word recognition.

195 1.3. *The current study*

Although it has been shown that individual differences in the capacity for statistical learning are linked with variability in word reading accuracy (Arciuli, 2018; Arciuli and Simpson, 2012) we still do not have a clear understanding of the mechanisms that give rise to these associations. What role do
200 domain-general mechanisms of implicit statistical learning play in extracting morphological structure and establishing sub-lexical morpho-orthographic to lexico-semantic mappings? To what extent do individual differences in implicit statistical learning impact this process? The first steps in answering these questions are to determine the extent to which differences in readers’
205 sensitivity to the internal structure of words is reflected in differences in the

patterns of brain responses to words that differ in morphological structure.

The goal of the study was to examine the extent to which individual differences in sensitivity to the internal structure of words is reflected in differences in patterns of brain responses. To examine potential differences
210 in sensitivity to morphological structure, we manipulated the cumulative root frequency, morphological family size and morphological complexity of non-word targets, and measured the amplitude of the N250 component as participants completed a lexical decision task.

Complex word recognition is influenced by many different kinds of lexical
215 and sub-lexical frequency information; four types of frequency information have been shown to be significantly influence response times—surface frequency, base frequency, cumulative root frequency and family size. Surface frequency (also known as token or lemma frequency) refers to the frequency of occurrence of a specific word form. Higher surface frequency words are
220 generally recognized faster than low frequency words—the word frequency effect.

Base frequency refers to the frequency of the base or root morpheme of a word, cumulative root frequency refers to the summed token frequency of all word forms containing a particular root and finally morphological family
225 size refers to the number (type frequency) of different words that share a common morphological root.

Research has shown that all other things being equal words with larger family sizes, higher surface frequencies, higher base frequencies and larger family sizes are often processed faster in visual lexical decision tasks. For
230 complex words a high base frequency can facilitate the recognition of complex

words sharing that base potentially because of the increased activation of related lexical entries. H

and morpheme frequency effects have often been used to test whether derived words activate morphemic representations;

235 The cumulative root frequency effect refers to the finding that words with high cumulative root frequency are responded to faster and more accurately than words with low cumulative root frequency when surface frequency is controlled (Taft, 1979) Taft (1979). The family size effects refers to the finding that lexical decision response times to words with larger family sizes (i.e., ap-
240 pearing as a constituent in larger numbers of derived words and compounds) are faster than for words with smaller family sizes. The cumulative root frequency effect only occurs in words with productive affixes, whereas the family size effect occurs regardless of affix productivity. Semantically transparent family members drove family size effects and the family size effect was
245 largely absent or attenuated in semantically opaque family members. Both family size and cumulative root frequency reflect sensitivity to morphological structure, but there is not consensus in the field regarding their relationship. The family size effect has hypothesized to reflect the activation of semantic representations as it sensitive to semantic transparency. In contrast, the cu-
250 mulative root frequency effect is affected by affix productivity, hence it has been hypothesized to the activation of morphological form representations.

The base frequency effect srefers to the influence of a word's root (or base) frequency on how quickly and easily we process morphologically complex words containing that root. In simpler terms, if a word shares a frequently
255 used root with other words, it tends to be processed faster than a similar

word with a less frequent root. This effect suggests that our brains might be sensitive to the morphological structure of words and how often we encounter those individual components.

In this study we chose to examine non-word processing in order to manipulate the orthographic properties of stimuli with greater precision than is possible using real words. We examined the effect of morphological structure on non-word recognition via the morpheme interference effect, or non-word complexity effect. This effect occurs when non-words that are created by combining existing morphemes (e.g., GASFUL) are rejected more slowly in a lexical decision task than words that do not have lexical structure (e.g., GAS-FIL). We hypothesized that the amplitude of the N250 component to these non-words would be modulated by participant reading style, as well as by morphological family size, and/or cumulative root frequency, and specifically that this modulation of the n250 response by morphological structure would be greater for readers with greater sensitivity to the morphological structure of words, or better ‘lexical quality’. Our dependent measure was the magnitude N250 response. The N250 is a negative-going wave that starts as early as 110 milliseconds and peaks around 250 milliseconds. Holcomb and Grainger (2006) suggest that the N250 reflects the process that maps sub-lexical orthographic and phonological representations onto whole word representations.

In masked morphological priming studies, the N250 has been found to be affected by semantic transparency, with unrelated items in transparent conditions showing significantly more negative responses than related items (Morris et al., 2007)

The study was designed to test the hypothesis that skilled readers engage in morpho-orthographic decomposition when processing visually presented non-words that resemble real morphologically complex words. Specifically, we predicted that ERP responses would reflect early decomposition processes for non-words that contain both a real stem and a real affix (e.g., gasful), but not for structurally similar non-words that contain a real stem and a non-affix (e.g., gasfil). These morphologically complex and simple non-words were matched in surface form but differed in whether they conformed to the structure of real morphologically derived words. If readers are sensitive to morpho-orthographic structure, we expected this contrast to elicit differences in ERP amplitude, particularly in the N250 time window, which has been linked to early form-based morphological processing. We further predicted that sensitivity to morpho-orthographic structure would be modulated by morphological family size—that is, the number of real words sharing the same stem. Because stems with large morphological families occur in more derived forms and thus provide stronger cues for decomposition, we expected an additive effect such that complex non-words with large-family stems would elicit the strongest evidence of decomposition. In contrast, simple non-words and items with small-family stems were expected to show reduced or absent decomposition effects. Finally, we predicted that individual differences in lexical quality—indexed by spelling and vocabulary measures—would modulate these effects, with stronger and more reliable decomposition effects observed in participants with higher sensitivity to orthographic structure.

2. Methods

305 2.1. *Participants*

We tested 75 participants recruited from the UMass–Five Colleges consortium community in Amherst, MA. Participants were included if they were between the ages of 18 and 26, right-handed, reported having normal or corrected-to-normal vision with no history of cognitive or reading-related
310 difficulties and were native English speakers with no other languages spoken in the home before the age of 3. Three participants were excluded for exceeding the protocol’s age range, two for being non-native English speakers and nine were excluded due to data loss, resulting in a final sample of 61 participants (age range: 18–25 years, $M = 19.7$). Of these, 45 identified as
315 female, 15 as male, and one did not report their sex. All participants provided informed consent prior to the start of the study and were paid for their participation. All materials and procedures were reviewed and approved by the IRB committee at Hampshire College.

To assess whether our study was adequately powered to detect effects of
320 similar magnitude to those reported in prior ERP studies ...

2.2. *Materials and Procedure*

The stimuli and procedure were adopted from

3. Results

4. Discussion

325 For cohort 1 the crossover patterns in the N250 window suggest that participants with higher orthographic sensitivity exhibit distinct neural process-

ing patterns consistent with early sensitivity to morpho-orthographic structure. For participants with a high sensitivity to morpho-orthographic structure, complex items evoked more negative amplitudes than simple items. For participants with low sensitivity, the direction reversed: complex items were somewhat less negative than simple items. With respect to morphological family size, for participants with low sensitivity to morpho-orthographic structure, items from large morphological families elicited more negative amplitudes than items from small families, but for high sensitivity participants, the direction reversed, with large family items evoking less negative amplitudes than small family items.

For cohort 2, ERP amplitudes were overall significantly less negative for complex items than for simple items. For items with small morphological families, simple non-words elicited significantly more negative ERP amplitudes than complex ones. Additionally for simple non-words, items with a small morphological family size evoked more negative amplitudes than those with large family size.

The N250 is associated with early form-based processing, so these effects suggest that morphological decomposition or structured form recognition is more active in high-sensitivity participants, whereas low-sensitivity participants show less structured engagement.

These results provide partial support for the hypothesis that ERP responses reflect sensitivity to morpho-orthographic structure. As predicted, morphological complexity modulated ERP amplitudes, particularly in the N250 time window associated with early form-based processing. In both cohorts, complex pseudowords—which contained real stems and real affixes—elicited

distinct neural responses compared to simple pseudowords with real stems and non-affixes, especially when morphological family size was small. This pattern is consistent with the notion that readers engage in morphological decomposition when presented with orthographically structured non-words that resemble real morphological combinations. Importantly, in Cohort 1, these effects were observed only in interaction with individual differences in orthographic sensitivity, suggesting that decomposition processes are more likely to be engaged by readers with higher-quality lexical representations. In Cohort 2, the effects of complexity and family size were more robust and less dependent on individual differences, but still aligned with the view that decomposition is more likely when the constituent stem belongs to a large morphological family. Critically, the N400 results in Cohort 2 further support this interpretation: greater negativity was observed for simple items with small family size, suggesting increased processing demands when decomposition cues were weak or misleading. The convergence of complexity and family size effects, especially in the N250 window, supports an additive model of decomposition, whereby items that are both morphologically complex and drawn from large families are more readily parsed. Together, these findings suggest that morpho-orthographic decomposition is an early, pre-lexical process that is modulated both by the structural properties of the stimulus and by reader-specific lexical knowledge.

References

Andrews, S., 2012. Individual differences in skilled visual word recognition and reading: The role of lexical quality, in: Adelman, J. (Ed.), Visual Word

Recognition: Meaning and Context, Individuals and Development, Vol. 2. Psychology Press, New York, NY, US. Current Issues in the Psychology of Language, pp. 151–172.

Andrews, S., Bond, R., 2009. Lexical expertise and reading skill: Bottom-up
380 and top-down processing of lexical ambiguity. *Reading and Writing* 22,
687–711. doi:10.1007/s11145-008-9137-7.

Andrews, S., Hersch, J., 2010. Lexical precision in skilled readers: Individual differences in masked neighbor priming. *Journal of Experimental Psychology: General* 139, 299–318. doi:10.1037/a0018366.

385 Andrews, S., Lo, S., 2013. Is morphological priming stronger for transparent than opaque words? It depends on individual differences in spelling and vocabulary. *Journal of Memory and Language* 68, 279–296. doi:10.1016/j.jml.2012.12.001.

Arciuli, J., 2018. Reading as statistical learning. *Language, Speech, and Hearing Services in Schools* 49, 634–643. doi:10.1044/2018_LSHSS-STLT1-17-0135.
390

Arciuli, J., Simpson, I.C., 2012. Statistical learning is related to reading ability in children and adults. *Cognitive Science* 36, 286–304. doi:10.1111/j.1551-6709.2011.01200.x.

395 Baayen, H., Milin, P., Durdevic, D.F., Hendrix, P., Marelli, M., 2011. An amorphous model for morphological processing in visual comprehension based on naive discriminative learning. *Psychological Review* 118, 438–481. doi:10.1037/a0023851.

- Castles, A., Rastle, K., Nation, K., 2018. Ending the reading wars: Reading
400 acquisition from novice to expert. *Psychological Science in the Public
Interest* 19, 5–51. doi:10.1177/1529100618772271.
- Cronbach, L.J., 1957. The two disciplines of scientific psychology. *American
Psychologist* 12, 671–684. doi:10.1037/h0043943.
- Feldman, L.B., O'Connor, P.A., Del Prado Martín, F.M., 2009. Early
405 morphological processing is morphosemantic and not simply morpho-
orthographic: A violation of form-then-meaning accounts of word recogni-
tion. *Psychonomic Bulletin & Review* 16, 684–691. doi:10.3758/PBR.16.
4.684.
- Forster, K.I., Davis, C., 1984. Repetition priming and frequency attenuation
410 in lexical access. *Journal of Experimental Psychology: Learning, Memory,
and Cognition* 10, 680–698.
- Forster, K.I., Taft, M., 1994. Bodies, antibodies, and neighborhood-density
effects in masked form priming. *Journal of Experimental Psychology:
Learning, Memory, & Cognition* 20, 844–863.
- 415 Frith, U., 2017. Beneath the Surface of Developmental Dyslexia, in: Patter-
son, K.E., Marshall, J.C., Coltheart, M. (Eds.), *Surface Dyslexia*. 1 ed..
Routledge, pp. 301–330. doi:10.4324/9781315108346-18.
- Holcomb, P.J., Grainger, J., 2006. On the time course of visual word recog-
nition: An event-related potential investigation using masked repetition
420 priming. *Journal of Cognitive Neuroscience* 18, 1631–1643.

Mahony, D., Singson, M., Mann, V., 2000. Reading ability and sensitivity to morphological relations. *Reading and Writing* 12, 191–218. doi:10.1023/A:1008136012492.

Perfetti, C.A., 1992. The representation problem in reading acquisition, in: Gough, P.B., Ehri, L.C., Treiman, R. (Eds.), *Reading Acquisition*. Lawrence Erlbaum Associates, Inc, Hillsdale, NJ, England, pp. 145–174.

Perfetti, C.A., 2007. Reading ability: Lexical quality to comprehension. *Scientific Studies of Reading* 11, 357–383. doi:10.1080/10888430701530730.

Perfetti, C.A., 2017. Lexical quality revisited, in: Segers, E., den Broek, P.v. (Eds.), *Developmental Perspectives in Written Language and Literacy: In Honor of Ludo Verhoeven*. John Benjamins, Amsterdam, Netherlands, pp. 51–67.

Perfetti, C.A., Hart, L., 2002. The lexical quality hypothesis, in: Verhoeven, L., Elbro, C., Reitsma, P. (Eds.), *Precursors of Functional Literacy*. John Benjamins Publishing. volume 11, pp. 67–86.

Plaut, D.C., Gonnerman, L.M., 2000. Are non-semantic morphological effects incompatible with a distributed connectionist approach to lexical processing? *Language and Cognitive Processes* 15, 445–485.

Rastle, K., 2019. The place of morphology in learning to read in English. *Cortex* 116, 45–54. doi:10.1016/j.cortex.2018.02.008.

Rastle, K., Davis, M.H., 2008. Morphological decomposition based on the analysis of orthography. *Language and Cognitive Processes* 23, 942–971.

- Rayner, K., Foorman, B.R., Perfetti, C.A., Pesetsky, D., Seidenberg, M.S.,
2001. How psychological science informs the teaching of reading. *Psycho-*
445 *logical Science in the Public Interest* 2, 31–74. doi:10.1111/1529-1006.
00004.
- Share, D.L., 1995. Phonological recoding and self-teaching: Sine qua non of
reading acquisition. *Cognition* 55, 151–218. doi:10.1016/0010-0277(94)
00645-2.
- 450 Taft, M., 1979. Recognition of affixed words and the word frequency effect.
Memory & Cognition 7, 263–272.
- Treiman, R., Kessler, B., 2006. Spelling as statistical learning: Using con-
sonantal context to spell vowels. *Journal of Educational Psychology* 98,
642–652. doi:10.1037/0022-0663.98.3.642.
- 455 Treiman, R., Mullennix, J., Bijeljac-Babic, R., Richmond-Welty, E.D., 1995.
The special role of rimes in the description, use, and acquisition of English
orthography. *Journal of Experimental Psychology: General* 124, 107–136.
doi:10.1037/0096-3445.124.2.107.
- Ziegler, J., Goswami, U., 2005. Reading acquisition, developmental dyslexia,
460 and skilled reading across languages: A psycholinguistic grain size theory.
Psychological Bulletin 131, 3–29.